CS 112

Introduction to Data Structures

WEEK 0

Intro to C++

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This Week

1. Structure of a C++ program
2. Compilation pipeline
3. Types, variables, I/O
4. Functions
5. Pointers **KEY**
6. Arrays
7. C++ vs. Python

Your First C++ Program

```cpp
int main() {


    return 0;
}
```

> Every C++ program needs a `main()` — this is where execution begins.

Your First C++ Program

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

Notice how `main()` morphed into its final form — we added the include, namespace, and output line.

Anatomy of a C++ Program

```cpp
#include <iostream>
using namespace std;

int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

1 `#include <iostream>`

Pulls in the I/O library — C++'s version of `import`.

2 `using namespace std;`

Lets you write `cout` instead of `std::cout`.

5 `cout << … << endl;`

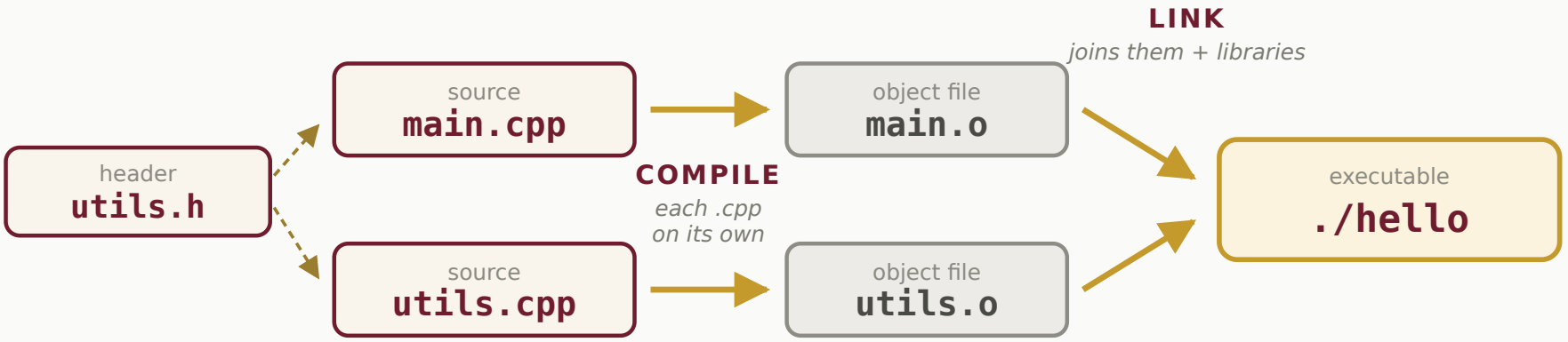
Prints to the terminal. `<<` is the "insertion" operator.

6 `return 0;`

Tells the OS the program finished cleanly.

Compilation Pipeline

headers are #included — pasted into each .cpp, never compiled on their own



COMPILE-TIME ERROR

```
error: 'cout' was not declared in
this scope
```

One file didn't make sense on its own — a typo, a missing `#include`, a type mismatch.

LINK-TIME ERROR

```
undefined reference to `helper()'
```

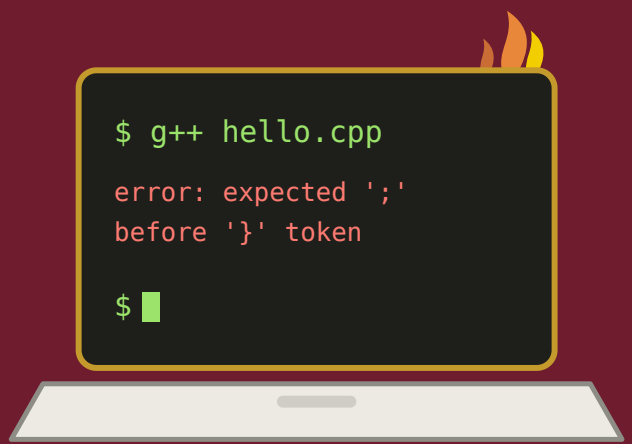
Every file compiled fine, but a function was *declared* and never *defined*.

Compilation: Key Points

- The compiler **translates** source code into machine-readable object files
- It also **type-checks** — many errors are caught before your program ever runs
- The linker **combines** object files and resolves references between them
- Libraries like `<iostream>` are pulled in during linking

> Unlike Python, you *must* compile before running. Upside: a whole class of bugs is caught at compile time, not in production.

Demo



Live coding. What could possibly go wrong?

(the semicolon is always missing on line 12)

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What does a compiler do?

A. Create an executable program from source code.

B. Check the source code for errors.

C. Pull in required libraries.

D. A and B only.

E. B and C only.

F. A, B, and C.

G. None of the above.

Data Types & Sizes

`int`	`unsigned`	`short`	`double`
4 bytes	4 bytes	2 bytes	8 bytes
−2B to 2B	*0 to 4B*	*−32K to 32K*	*15–16 digits*

`float`	`bool`	`char`	`string`
4 bytes	1 byte	1 byte	varies
6–7 digits	*true / false*	*'A', '\n'*	*"hello"*

```
cout << sizeof(int);       // 4   — check any type yourself
cout << sizeof(double);    // 8
cout << sizeof(grade);     // 1   — works on variables too
```

Types are **fixed at compile time**. Sizes can vary by machine — `sizeof` tells you the truth on yours.

Python vs. C++: Variables

Python

```
x = 7
```

namespace

x → **obj @ 0xABCD**

object space

0xABCD **int, value 7**

`x` is a *reference* to a typed object.

C++

```
int x = 7;
```

memory

0x12345678 **7** ← x

(4 bytes)

`x` *is* the memory location — it holds the value directly.

Declare, Then Initialize

```
int    count;              // declared — NOT initialized
int    total  = 0;         // declared and initialized
double pi     = 3.14159;
char   grade  = 'A';
bool   ready  = true;
string name   = "Ada";

cout << count;             // ⚠ undefined — could print anything
```

```
0x7ffd2a10  -8593204  ← count  (leftover bytes)
0x7ffd2a14      0     ← total  (you put it there)
```

In Python a name doesn't exist until you assign to it. In C++ the box exists the moment you *declare* it — holding whatever the last program left in that memory.

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Will this C++ code compile?

```
int x = 11;
x = "12";
```

A. Yes

B. No

C. Yes, but with warnings.

Using const

```
const int    MAX_SIZE = 100;
const double PI       = 3.14159265;

MAX_SIZE = 200;  // ❌ compiler error — read-only!
```

- Put `const` before the type to make a variable read-only
- The compiler **rejects** any code that tries to change it
- Convention: use `ALL_CAPS` names for constants

> Use `const` liberally — it documents intent and catches bugs at compile time.

Input / Output

```cpp
#include <iostream>
using namespace std;

int main() {
    double value;
    cout << "Enter a number: ";   // output → terminal
    cin  >> value;                // input  ← terminal
    cerr << "got " << value << endl; // error stream

    cout << "You entered: " << value << endl;
    return 0;
}
```

`cout` + `<<` — "insertion" (output to terminal)

`cin` + `>>` — "extraction" (input from terminal)

File I/O

```cpp
#include <iostream>
#include <fstream>
#include <cmath>
#include <cassert>
using namespace std;

int main() {
    ifstream fin("numbers.txt");
    assert(fin.is_open());   // stop if the file is missing

    double n;
    while (fin >> n) {       // false at end of file
        cout << sqrt(n) << endl;
    }
    fin.close();
    return 0;
}
```

`ifstream` reads from a file just like `cin` reads from the terminal. Use `ofstream` to write.

Decimal, Binary, Hexadecimal

```
int x = 33;         // decimal  (default)
int y = 0x21;       // hex:    2×16 + 1 = 33
int z = 0b100001;   // binary: 1×32 + 1 = 33

cout << x << " " << y << " " << z << endl;
// Output:  33 33 33
```

- `0x` prefix → hexadecimal (base 16)
- `0b` prefix → binary (base 2)
- Memory addresses are typically displayed in hex

Three ways to write the *same number*. The compiler stores them identically in memory.

Why bother with binary?

Everything on the machine is bits. Hex is just a friendlier way to read them.

Functions

Write the tests *before* the function — it forces you to think about what it should do:

```
assert(factorial(0) == 1);
assert(factorial(1) == 1);
assert(factorial(4) == 24);
assert(factorial(5) == 120);
```

This test-first style is called **test-driven development**. Your assertions become the spec.

Functions

```
unsigned factorial(unsigned n) {
    unsigned result = 1;
    for (unsigned i = 2; i <= n; ++i) {
        result *= i;
    }
    return result;
}

int main() {
    assert(factorial(0) == 1);
    assert(factorial(1) == 1);
    assert(factorial(4) == 24);
    assert(factorial(5) == 120);
    cout << "All tests passed!" << endl;
    return 0;
}
```

Pointers

A pointer is a variable that holds a **memory address**.

```
int y = 77;
```

0x12345678	**77**	← y

Pointers

Use `&` ("address-of") to get a variable's address:

```
int y    = 77;
int *x   = &y;   // x holds the address of y
```

0x12345678	**77**	← y
0x12345690	**0x12345678**	← x (pointer to y)

`int *x` declares x as a "pointer to int". Its value is an address, not 77.

Pointers: Dereferencing

Use `*` to get (or set) the value *at* the address:

```
int y  = 77;
int *x = &y;

cout << x;     // 0x12345678  — the address
cout << *x;    // 77          — follow the pointer
*x = 99;       // write through the pointer
cout << y;     // 99          — y itself changed
```

0x12345678	**99**	← y (changed via *x)
0x12345690	**0x12345678**	← x

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```
int y = 77;    // placed at address 0x12345678
int *x = &y;  // placed at address 0x12345690
cout << x;
```

What is printed?

A. 77

B. 0x12345678

C. 0x12345690

D. Error

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```
int y = 77;
int *x = &y;
cout << *y;   // note: *y, not *x
```

What is printed?

A. 77

B. 0x12345678

C. 0x12345690

D. Compile error

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```
int y = 77;
int *x = &y;
cout << *x;
```

What is printed?

A. 77

B. 0x12345678

C. 0x12345690

D. Error

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```
int y = 77;
int *x = &y;
*x = 78;
cout << *x;
```

What is printed?

A. 77

B. 78

C. 0x12345690

D. Error

TALK TO YOUR NEIGHBOR · TTYN

```
int y = 77;
int *x = &y;
*x = 78;
cout << y;    // note: y, not *x
```

What is printed?

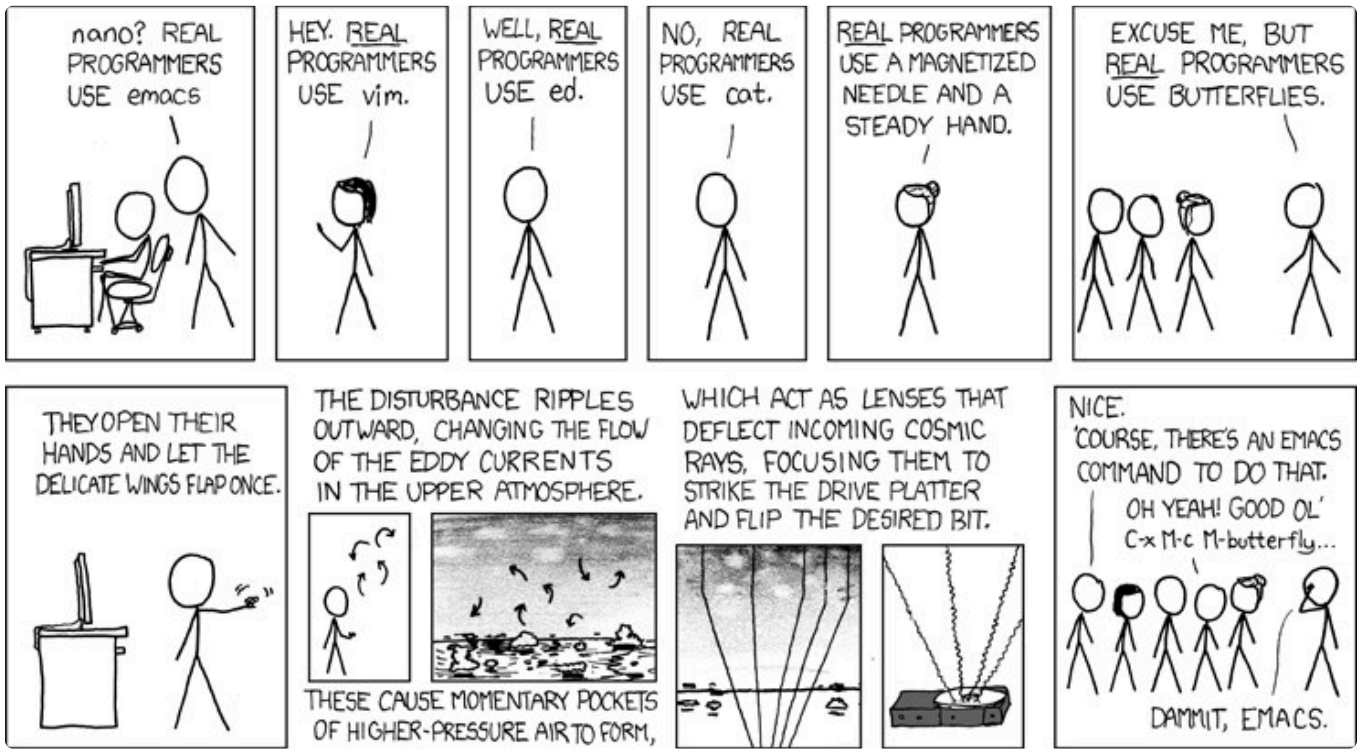
A. 77

B. 78

C. 0x12345690

D. Error

Meanwhile, in the C++ community...



You just spent five slides flipping bits by hand. Some people skip the pointer.

Arrays

Like Python lists, but **fixed size** and **no bounds checking**.

```
char  letters[26];              // 26 chars, uninitialized
float scores[10] = { 0 };       // all 0.0
int   vals[] = { 10, 9, 4, 2, 0 };  // size inferred: 5
```

- Size must be known at compile time (for stack arrays)
- Indexes: `0` to `n-1`
- Accessing out of bounds → **undefined behavior**, no crash guaranteed

Arrays: Memory Layout

```
int arr[6] = { 1, 3, 5, 2, 77, -11 };
```

All elements are **contiguous** in memory — no gaps:

index	[0]	[1]	[2]	[3]	[4]	[5]
base+0	**1**	**3**	**5**	**2**	**77**	**-11**

> The array name `arr` is a pointer to the first element — this links arrays and pointers directly.

Pointers & Arrays

```
int arr[] = { 3, -11, 4, 12 };
int *ptr  = &arr[2];   // ptr points to arr[2] → value 4

cout << *ptr;          // 4
cout << *(ptr + 1);    // 12   (next element)
cout << *(ptr - 1);    // -11  (previous element)
```

Pointer arithmetic lets you navigate arrays by address. This is how arrays *actually work* under the hood in C++. More on this in upcoming weeks.

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```
int arr[] = { 3, -11, 4, 12 };
int *ptr  = &arr[2];
cout << *ptr;
```

What is printed?

A. 3

B. 2

C. 4

D. -11

C++ vs. Python: Core Differences

Feature	Python	C++
Variables	Reference to a typed object	Named memory location with explicit type
Type checking	Dynamic (runtime)	Static (compile time)
Function return type	Implied from `return`	Declared explicitly
1D data structure	`list`	`array` or `vector`
Memory management	Automatic (garbage collected)	Mostly explicit
Entry point	First line executed	`main()`

C++ vs. Python: More Differences

Feature	Python	C++
Interface / impl split	Not explicit	`.h` header + `.cpp` source
Parameter passing	Pass-by-value (objects by ref)	By-value, by-reference, by-const-ref
Access control	Mostly none	`public` / `private`
Constructors	Called explicitly	Explicit *or implicit* (scope entry)
Destructors	None (GC handles it)	Defined; called implicitly at scope exit
Generics	None	Templates — powerful but complex
Strings	Built-in	`#include <string>`

Week 0 Recap

- C++ programs need a `main()` and must be **compiled** before running
- Types are **explicit and checked at compile time**
- Pointers hold **memory addresses** — the key concept in C++
- Arrays are contiguous memory blocks — fast, but no safety net
- C++ gives you more control than Python, and more responsibility

Next: Getting Started (pre-lab) · then a01 — RPG Character Creator →

This deck stands on earlier CS112 materials by
Joel Adams and **Victor Norman** · adapted and extended by **Eric Araújo**